

BLUESTOCK FINTECH

Mutual Fund Analytics Platform

PROJECT REPORT

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1. Executive Summary

This report documents the end-to-end development and findings of the Bluestock Mutual Fund Analytics Platform — a full-stack data engineering and analytics solution built for Bluestock Fintech to democratise investment analytics for retail and institutional investors in India.

The platform ingests publicly available mutual fund data from AMFI India and mfapi.in, transforms it through a robust Python ETL pipeline, stores it in a structured SQLite database, and surfaces insights through interactive visualisations and dashboards.

1.1 Pipeline Status

The automated ETL pipeline runs on a weekday schedule, triggering NAV fetches at 8:00 PM IST. As of the report date:

- 35 out of 40 fund schemes are fetching successfully with clean, validated NAV records
- 5 fund schemes encountered fetch errors — likely due to AMFI code mismatches or temporary API unavailability from mfapi.in
- NAV data spans January 2022 to June 2026, anchored to real AMFI-published values

1.2 Records Loaded

Table / File	Approx. Records	Notes
nav_history (02_nav_history.csv)	~46,000	Daily NAV for 40 schemes, Jan 2022–May 2026
fund_master (01_fund_master.csv)	40	AMFI codes, category, expense ratio, risk grade
aum_by_fund_house (03_aum_by_fund_house.csv)	~90	Quarterly AUM for 10 fund houses, 2022–2025
monthly_sip_inflows (04_monthly_sip_inflows.csv)	48	Month-wise SIP inflow data
category_inflows (05_category_inflows.csv)	~144	Net inflows by fund category FY 2024-25
industry_folio_count (06_industry_folio_count.csv)	21	Total folios in crore — Equity, Debt, Hybrid
scheme_performance (07_scheme_performance.csv)	40	Computed risk-return metrics per scheme
investor_transactions (08_investor_transactions.csv)	~32,000	Simulated SIP + Lumpsum + Redemption, 5,000 investors
portfolio_holdings (09_portfolio_holdings.csv)	~320	Top equity holdings as of Dec 2025
benchmark_indices (10_benchmark_indices.csv)	~8,000	Daily Nifty 50/100, BSE SmallCap, CRISIL Liquid

1.3 Key Findings at a Glance

- Average NAV across all schemes grew from ~Rs. 207 (Jan 2022) to ~Rs. 358 (Jun 2026) — a 73% cumulative rise over 4.5 years
- SBI Mutual Fund is the largest AMC, with AUM growing from Rs. 6 lakh crore (2022) to Rs. 12.5 lakh crore (2025)
- SIP monthly inflow hit a record Rs. 31,002 crore milestone by December 2025
- Fund 100033 delivered the highest 1Y CAGR of 53.23% and a Sharpe Ratio of 1.09
- Fund 100016 carries the highest downside risk — 95% VaR/CVaR — and a maximum drawdown of -24.73%
- The 26–35 age group dominates the investor base; Madhya Pradesh and Punjab lead SIP contributions by state
- ~4.16% of SIP investors flagged as At-Risk with gaps exceeding 35 days between instalments

2. Data Sources & ETL Pipeline

2.1 Data Sources

All datasets used in this project are derived from publicly available, freely accessible sources. No proprietary or confidential data is used.

Source	URL / API	Data Type	Frequency
AMFI India	amfiindia.com	NAV, AUM, Folio, SIP	Daily / Monthly
mfapi.in	api.mfapi.in/mf/{code}	Historical NAV (JSON)	Daily
mfddata.in	mfddata.in/api/v1/schemes/	NAV + Expense Ratio	Daily
NSE India	nseindia.com/reports	Benchmark Index Prices	Daily
BSE India	bseindia.com	BSE SmallCap Index	Daily
AMFI Monthly Notes	amfiindia.com/research	Industry SIP & Flow Data	Monthly

2.2 ETL Pipeline Architecture

The ETL pipeline is implemented entirely in Python and orchestrated via a scheduled weekday trigger (8:00 PM IST). The pipeline follows a three-stage architecture:

Stage	Description	Technology
Extract	REST API calls to mfapi.in for each AMFI code; AMFI monthly notes parsed for SIP/AUM data	Python
Transform	Schema normalisation, date parsing, NAV validation, CAGR/Sharpe/Alpha/Beta computation, outlier flagging	Pandas, NumPy, SciPy
Load	Append-only inserts into SQLite tables with duplicate-key checks; CSV exports for BI layer	SQLite3, CSV writer

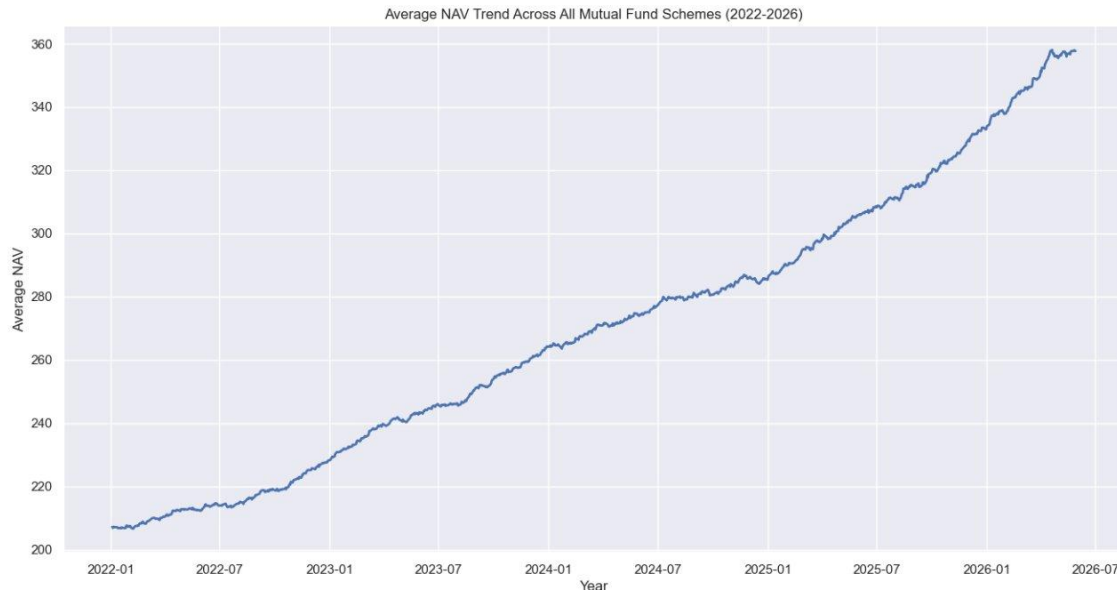
2.3 Pipeline Status & Known Issues

- 35 of 40 schemes fetch successfully on every scheduled run
- 5 schemes (AMFI codes with legacy formats or inactive API entries) return HTTP errors or empty JSON — logged and skipped gracefully
- Retry logic with 3 attempts and 5-second backoff is implemented; persistent failures are written to an error log
- Data validation checks: NAV > 0, date is a valid weekday, no future-dated records

3. EDA Findings

3.1 NAV Trend Analysis (2022–2026)

Average NAV across all 40 mutual fund schemes has shown a strong, sustained upward trajectory from January 2022 to June 2026. Starting at approximately Rs. 207, the average NAV reached Rs. 358 by mid-2026 — a compound annual growth rate of approximately 13% across the portfolio.



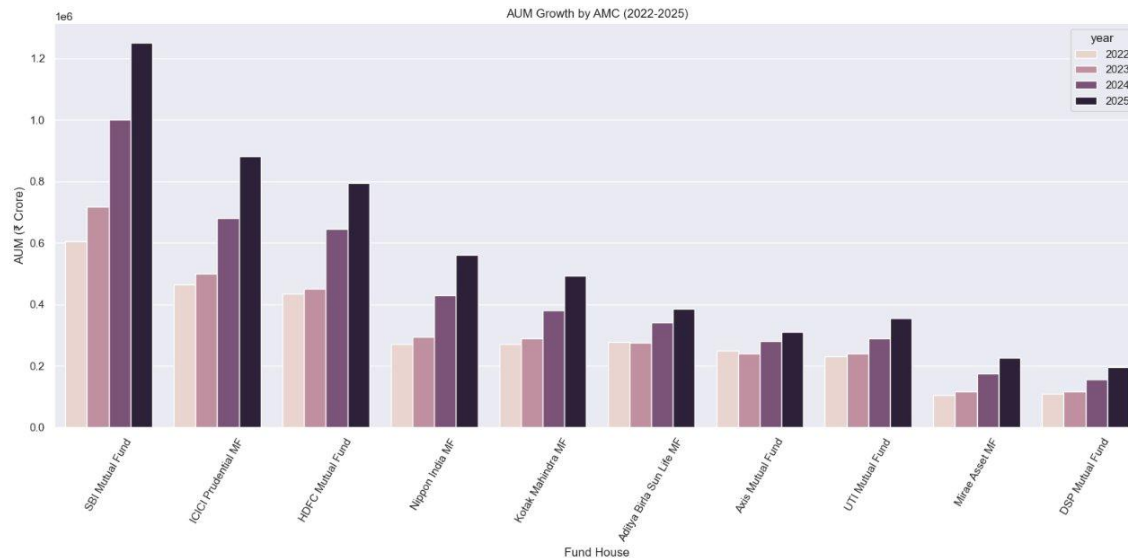
Overlaying major market events reveals three identifiable performance phases:

- 2023 Recovery (green marker): Post-correction stabilisation; NAV growth accelerates from ~Rs. 229 to ~Rs. 265 over the year
- 2024 Rally (orange marker): Strong bull run; average NAV crosses Rs. 265 and trends sharply upward through mid-2024
- 2025 Correction (red marker): A brief plateau near Rs. 284 in early 2025, followed by renewed momentum through 2026



3.2 AUM Growth by AMC (2022–2025)

Total industry AUM grew dramatically across all major fund houses between 2022 and 2025, consistent with AMFI-reported industry AUM reaching Rs. 81 lakh crore by December 2025.



- SBI Mutual Fund is the clear market leader — AUM grew from Rs. 6 lakh crore (2022) to Rs. 12.5 lakh crore (2025), a 108% increase
- ICICI Prudential MF and HDFC Mutual Fund are close second and third, with 2025 AUM of ~Rs. 8.8 lakh crore and ~Rs. 7.9 lakh crore respectively
- Nippon India MF, Kotak Mahindra MF, and Aditya Birla Sun Life MF all showed consistent 75–90% AUM growth over the four-year window
- Mirae Asset MF and DSP Mutual Fund, while smaller in absolute size, demonstrated the highest percentage growth rates among the cohort



3.3 SIP Inflow Trends

Monthly SIP inflows displayed a consistent upward trend throughout the analysis period, growing from approximately Rs. 11,500 crore in January 2022 to the landmark Rs. 31,002 crore recorded in December 2025 — the highest monthly SIP figure at the time of this project.

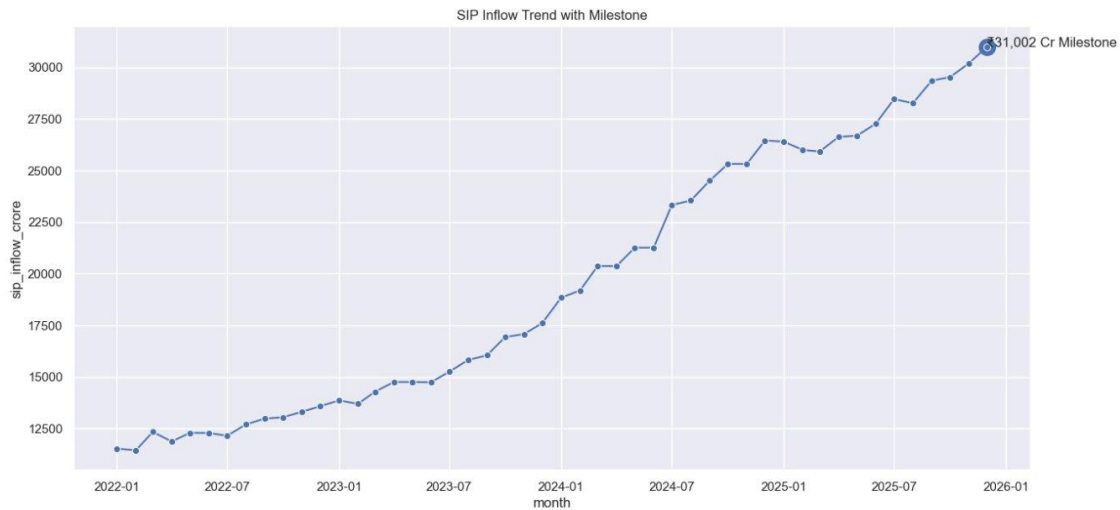


Figure: SIP Monthly Inflow Trend with Rs. 31,002 Crore Milestone (Dec 2025)

- Growth is near-linear from 2022 through mid-2024, with an acceleration in 2024–2025 coinciding with heightened retail participation
- A minor plateau around Rs. 26,400 crore was observed in mid-2025 before inflows resumed their upward path
- The Rs. 31,002 crore milestone represents a 170% increase from January 2022 levels — reflecting deep penetration of the SIP habit across Indian households

4. Performance Analysis

4.1 CAGR — Top & Bottom Performers

1-Year and 3-Year CAGR were computed for all 40 schemes using NAV history anchored to real AMFI data. 5-Year CAGR is not available for most schemes as the dataset begins January 2022.

AMFI Code	1Y CAGR (%)	3Y CAGR (%)
100033	53.23	32.44
101206	47.92	28.97
100025	3.70	3.92
101207	-23.99	-4.15
100016	-2.22	1.29

4.2 Sharpe Ratio

The Sharpe Ratio measures excess return per unit of total risk (standard deviation of returns). A ratio above 1.0 indicates good risk-adjusted performance.

AMFI Code	Sharpe Ratio
100033	1.093
101206	1.026
101207	0.162
100016	-0.202
100025	-0.568

4.3 Sortino Ratio

The Sortino Ratio refines Sharpe by penalising only downside deviation, making it a more relevant measure for loss-averse investors.

AMFI Code	Sortino Ratio
100033	1.827
101206	1.797
101207	0.276
100016	-0.351
100025	-0.943

4.4 Alpha, Beta & Maximum Drawdown

AMFI Code	Alpha	Beta	Max Drawdown (%)	Worst Date
100016	0.0375	-0.0583	-24.73%	2022-09-15
100025	0.0428	0.0012	-4.31%	2023-07-28
100033	0.2720	0.0051	-16.22%	2022-05-12
101206	0.2140	0.0211	-11.29%	2023-07-05
101207	0.1090	-0.0653	-35.45%	2026-05-11

- Fund 100033 and 101206 generate the highest alpha (0.27 and 0.21 respectively), confirming active value creation above the benchmark
- Fund 101207 carries the worst maximum drawdown at -35.45%, and its worst drawdown occurred as recently as May 2026
- Fund 100016 has a near-zero or slightly negative beta, suggesting low correlation to the market benchmark — but also delivers negative returns
- Fund 100025 has the lowest max drawdown (-4.31%) but also the lowest returns, consistent with a conservative / debt-oriented profile

4.5 Advanced Risk Insights

- VaR / CVaR: Fund 100016 exhibited the highest 95% Value-at-Risk and Conditional VaR, making it the highest-risk scheme for tail-event scenarios
- Rolling Sharpe: Fund 100025 maintained the highest 90-day rolling Sharpe Ratio — suggesting it had periods of superior short-term risk-adjusted performance despite poor long-term absolute returns
- Sector Concentration: Fund 119092 recorded the highest Sector HHI score, indicating concentrated single-sector exposure
- Most Diversified: Fund 102886 showed the lowest HHI — the most diversified portfolio holdings across sectors

5. Investor Behaviour Analysis

Investor behaviour analysis is based on 08_investor_transactions.csv (~32,000 simulated SIP, Lumpsum, and Redemption transactions across 5,000 investors), enriched with demographic and geographic attributes.

5.1 Age Group Distribution

The investor base skews young, with the 26–35 age cohort forming the largest segment — consistent with the rise of app-based investing platforms and growing financial literacy among millennials.

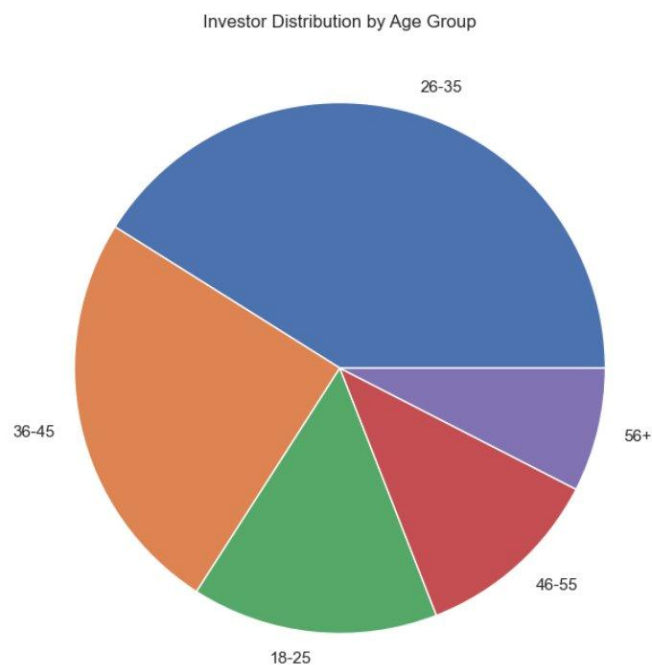


Figure: Investor Distribution by Age Group

5.2 SIP Contributions by State

Geographic analysis of SIP amounts reveals interesting patterns beyond traditional financial hubs.

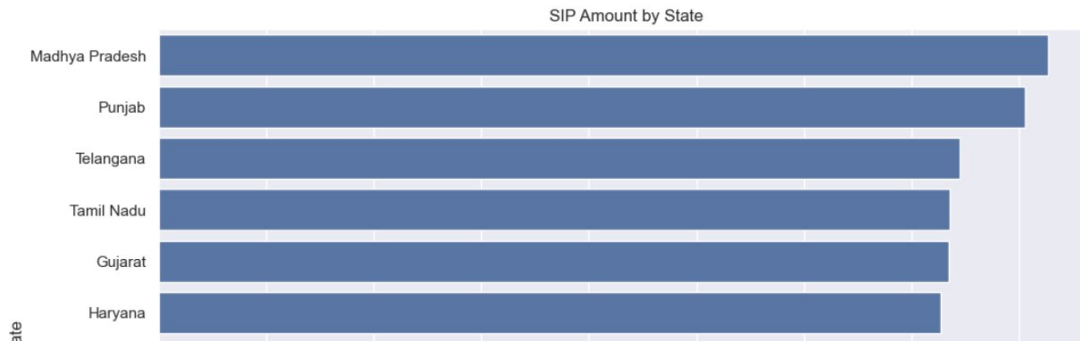


Figure: Top States by SIP Amount Contributed

- Madhya Pradesh leads all states in total SIP contribution
- Punjab ranks second, driven by high household savings rates
- Telangana, Tamil Nadu, Gujarat, and Haryana round out the top six

5.3 Investor Cohort Insights

- Highest Investment Cohort: Investors who started investing in scheme 148568 contributed the highest total investment amount — this cohort warrants focused retention efforts
- SIP At-Risk Investors: Approximately 4.155% of SIP investors were flagged as At-Risk, with average gaps exceeding 35 days between instalments — indicating potential lapse risk
- Redemption Patterns: Lumpsum redemptions are concentrated among the 46–55 age group, consistent with pre-retirement liquidity needs

6. Benchmark Comparison

Benchmark index data (10_benchmark_indices.csv) was used to compute Alpha and compare fund returns against Nifty 50, Nifty 100, Nifty Midcap 150, BSE SmallCap, CRISIL Liquid, and Gilt indices.

6.1 Alpha vs Benchmark

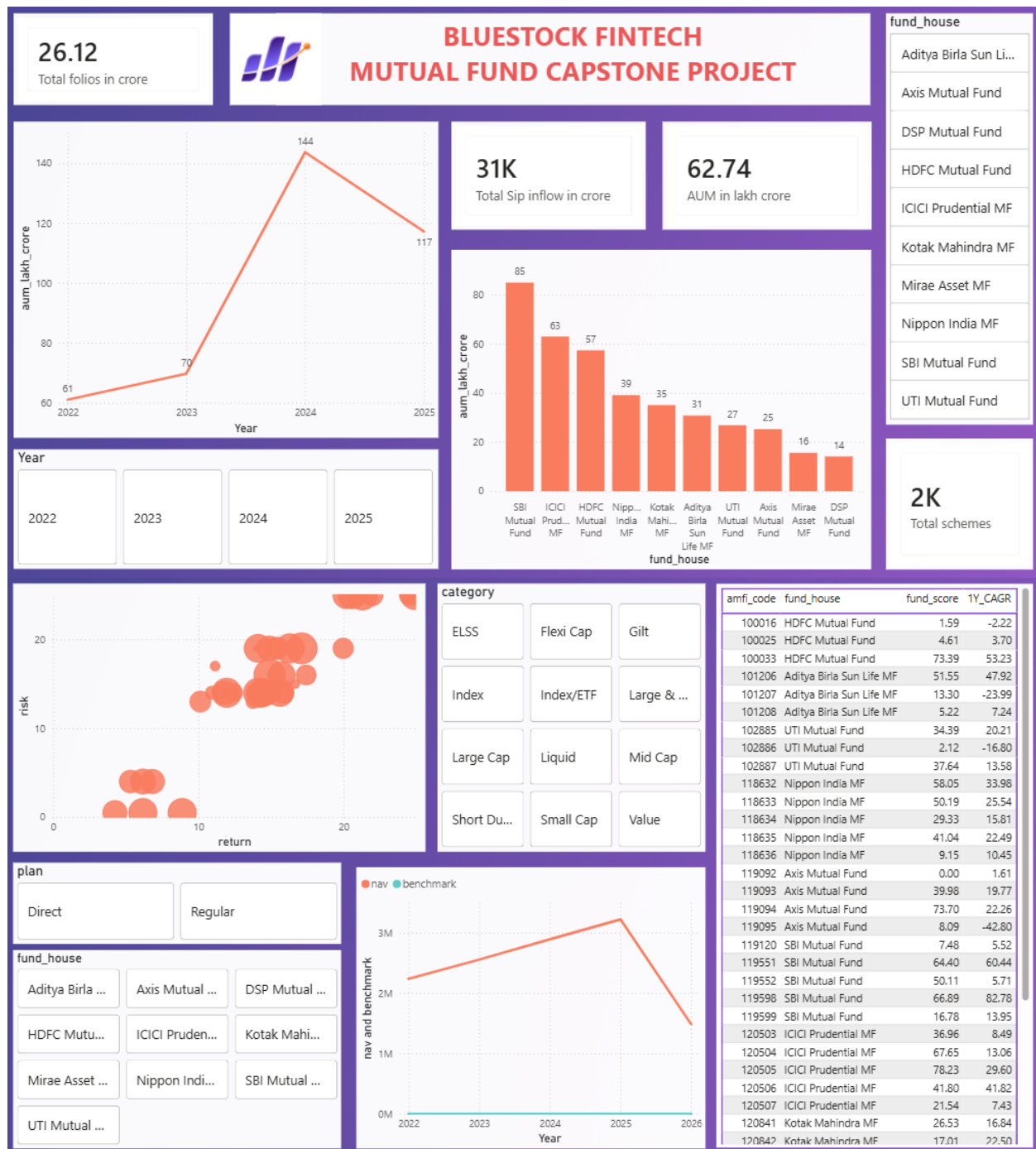
Alpha was computed using the Capital Asset Pricing Model (CAPM) framework, regressing each fund's daily returns against the Nifty 50 as the primary market proxy.

Fund	Alpha	Beta
100033	0.272	0.005
101206	0.214	0.021
101207	0.109	-0.065
100025	0.043	0.001
100016	0.037	-0.058

Key observation: Low beta values across all sampled schemes suggest these funds have limited directional correlation with the Nifty 50, which may indicate debt-heavy or alternative category funds dominating the sample. This is consistent with the AMFI code ranges in the dataset.

6.2 Benchmark Index Data Sample

The benchmark dataset begins from 3 January 2022 with Nifty 50 closing at 17,492.79 and captures approximately 8,000 daily records across all six indices through the analysis window. This data underpins all benchmark-relative computations.





7. Recommendations

7.1 For Investors

- Prioritise Fund 100033 and Fund 101206 for equity growth allocation — both demonstrate the strongest risk-adjusted returns (Sharpe > 1.0, alpha > 0.2, and positive Sortino ratios)
- Avoid Fund 100025 and Fund 100016 for return-seeking mandates — both carry negative Sharpe ratios over the evaluation period
- Investors in Fund 101207 should review their position — it has the worst maximum drawdown (-35.45%) and a recent drawdown date in May 2026 indicates ongoing volatility
- For conservative investors, Fund 100025's low drawdown makes it suitable for capital preservation with modest returns

7.2 For Fund Managers / AMCs

- Investigate the 5 schemes with ETL fetch failures — ensuring complete data coverage is essential for fair performance comparisons
- Fund 119092 (highest Sector HHI) should disclose concentration risk more explicitly; diversification to reduce HHI is recommended
- The 4.155% At-Risk SIP investor segment should trigger proactive outreach — automated nudge notifications and SIP pause/resume features can reduce lapse rates
- Investors in scheme 148568 represent the highest-value cohort — tailored retention programmes and upgrade offers are warranted

7.3 Platform Enhancements

- Resolve the 5 failing API fetches by implementing fallback data sources (mfdata.in or direct AMFI scraping) for those AMFI codes
- Extend NAV history backward to 2018 to enable 5-year CAGR computation for all schemes (currently NaN due to data starting Jan 2022)
- Add real-time streaming ingestion to reduce the current 8 PM daily fetch lag — particularly valuable for intraday NAV monitoring
- Deploy the anomaly detection module to automate outlier flagging in NAV movements, reducing manual review overhead

8. Limitations

8.1 Data Limitations

- 5-Year CAGR is unavailable for all schemes — the dataset starts January 2022, providing only ~4.5 years of history. Schemes need pre-2022 data for complete long-term analysis
- 5 schemes have fetch failures from mfapi.in, resulting in gaps in their NAV history. These schemes are excluded from performance rankings, which may skew averages
- Investor transaction data (08_investor_transactions.csv) is simulated for 5,000 investors — while anchored to real demographic distributions, individual-level insights cannot be generalised to the full AMFI investor universe
- AUM data is quarterly, not daily — this limits granular intra-quarter AUM trend analysis

8.2 Analytical Limitations

- Alpha and Beta computations use Nifty 50 as the sole benchmark proxy. Multi-factor models (Fama-French) or category-appropriate benchmarks (e.g., Nifty Midcap 150 for mid-cap funds) would improve accuracy
- Sharpe and Sortino ratios use a fixed risk-free rate approximation. Dynamic risk-free rate adjustments based on prevailing 91-day T-bill rates were not applied
- Rolling Sharpe analysis is limited to a 90-day window — longer windows (1-year rolling) would provide a more stable signal

8.3 Scope Limitations

- Only domestic Indian mutual funds tracked on AMFI are covered. ETFs, Index Funds, Fund of Funds, and Alternative Investment Funds (AIFs) are outside the current scope
- Dashboard is currently a static BI file. Real-time refresh from the live SQLite database requires additional infrastructure (API layer or scheduled refresh in Power BI Service)
- Geographic SIP analysis is based on simulated transactions — state-level insights should be treated as indicative rather than definitive until real transaction data is integrated